UBT837: CANCER BIOLOGY

L	T	P	Cr
3	0	0	3.0

Course Objective: The objective of this course is to introduce current concepts and advances in the area of cancer biology. The students will understand the role of oncogenes and suppressor genes and get knowledge on cancer related mutagens and pathways and cancer therapy.

Syllabus

Fundamentals of cancer biology: Regulation of cell cycle, mutations that cause changes in signal molecules, effects on receptor, signal switches, tumor suppressor genes, modulation of cell cycle in cancer, different forms of cancers, diet and cancer, cancer and chronic diseases.

Principles of carcinogenesis: Theory of carcinogenesis, Chemical carcinogenesis, metabolism of carcinogenesis, principles of physical carcinogenesis, x-ray radiation-mechanisms of radiation carcinogenesis.

Principles of molecular cell biology of cancer: Signal targets and cancer, activation of kinases; Oncogenes, identification of oncogenes, retroviruses and oncogenes, detection of oncogenes. Oncogenes/proto-oncogene activity. Growth factors related to transformation. Telomerases.

Principles of cancer metastasis: Clinical significances of invasion, heterogeneity of metastatic phenotype, metastatic cascade, basement membrane disruption, three step theory of invasion, proteinases and tumor cell invasion.

Cancer diagnostic and therapy: Cancer screening and early detection, Detection using biochemical assays, tumor markers, molecular tools for early diagnosis of cancer. Different forms of therapy, chemotherapy, radiation therapy, chemotherapy, detection of cancers, prediction of aggressiveness of cancer, advances in cancer detection.

Laboratory Work (if applicable)

Course Learning Objectives (CLO)
The students will be able to:

1. Comprehend pathogenesis, molecular mechanisms,
2. Identify cancer related risk factors
3. Explain cancer metastasis microenvironment and cancer therapy

4. Understand cancer diagnostic and therapeutic

Text Books

1. Robert A. Weinberg. The Biology of Cancer. 2nd edition, 2006.
2. Robin Hesketh. Introduction to Cancer Biology Cambridge, University Press 2013.
3. David J. Kerr, Daniel G. Haller, Cornelis J.H. van de Velde, Michael Baumann. The Oxford Textbook of Oncology. 3rd edition, 2016.

Reference Books

1. An Introduction Top Cellular and Molecular Biology of Cancer", Oxford Medical Publications, 1991.
2. Stella Pelengaris and Michael Khan. The Molecular Biology of Cancer, 2nd edition. Wiley Blackwell, 2013

Evaluation Scheme:

Sr. No.	**Evaluation Elements**	**Weightage (%)**
1.	MST	25-35
2.	EST	35-45
3.	Sessionals (May include assignments/quizzes)	30